ParkMate

A robotic assistant for parking cars

CSE461 – Introduction to Robotics
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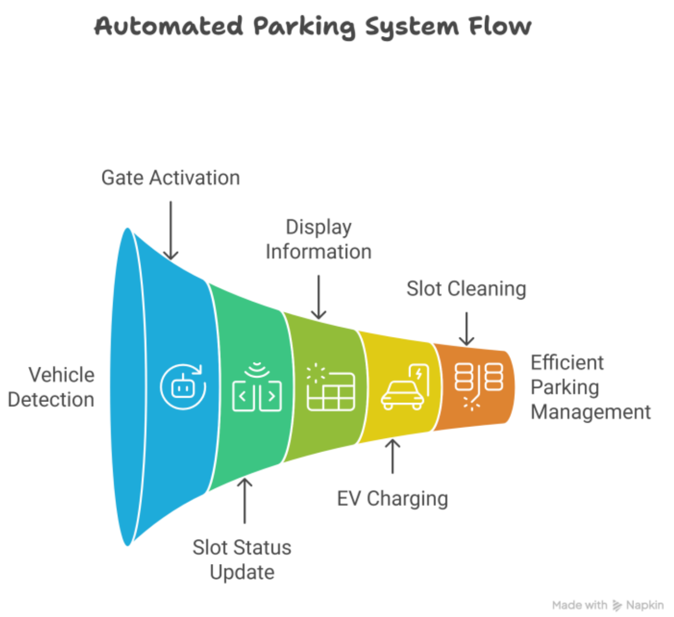
Abstract

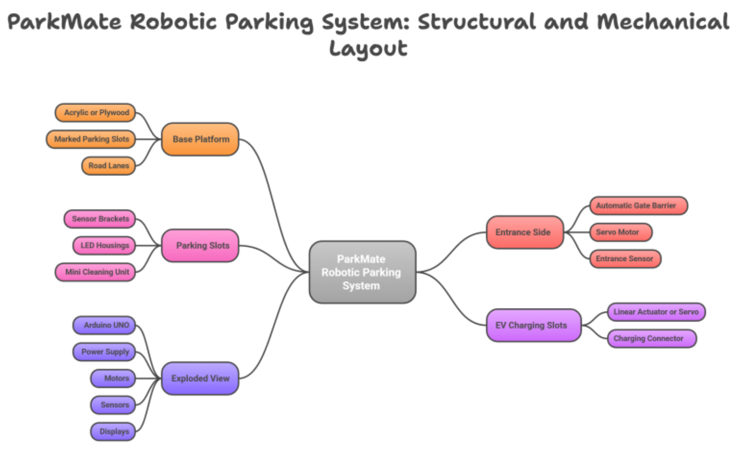
ParkMate is a smart robotic parking system designed to make parking easier, faster, and safer. It automatically detects incoming vehicles, opens the gate, and guides drivers to available parking slots using sensors and visual indicators. The system monitors each slot in real time, displays available spaces, and helps reduce confusion and traffic inside parking areas. For electric vehicles, ParkMate also offers automated charging in dedicated slots. By using simple robotics, sensors, and an Arduino controller, the system minimizes human effort, prevents parking mistakes, and saves time. ParkMate is suitable for malls, offices, residential areas, and small parking lots, showing how robotics can solve everyday urban parking problems in a smart and organized way.

Introduction

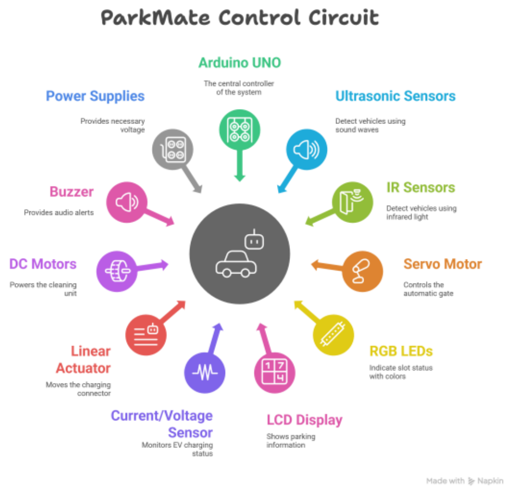
With the rapid growth of vehicles in urban areas, parking has become time-consuming, confusing, and inefficient. Drivers often struggle to find available spaces, leading to traffic congestion and wasted fuel. ParkMate is introduced to solve this problem by automating the parking process. The robot detects vehicles, manages entry, guides cars to free slots, and monitors parking in real time. It also supports electric vehicle charging. ParkMate aims to make parking faster, safer, and more organized, offering a practical robotic solution for modern parking spaces.

System Overview

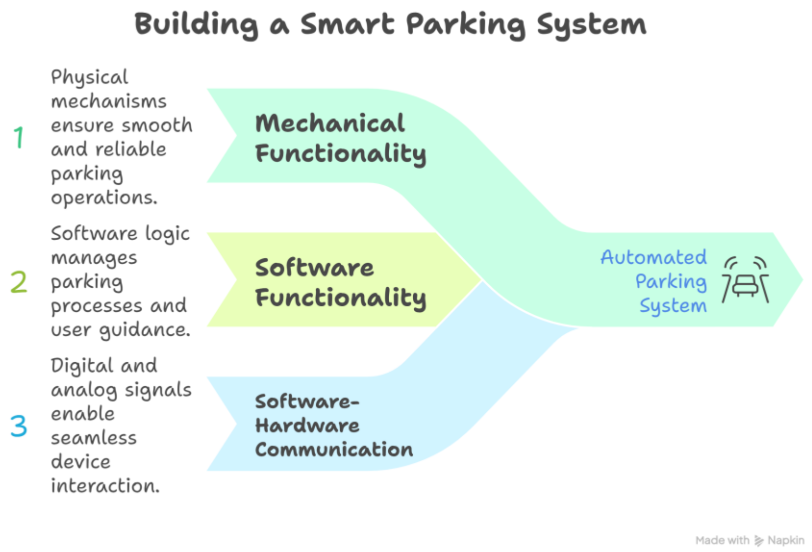




Implementation



Functionality



Applications and Competitors

Conclusion

In conclusion, ParkMate demonstrates how robotics and automation can transform traditional parking systems into smart, efficient, and user-friendly solutions. By integrating sensors, control systems, and mechanical actuation, the project reduces human effort, saves time, and improves safety. ParkMate highlights the practical use of robotics in solving real-world urban parking challenges.